Department of Electronic & Electrical Engineering
Faculty of Engineering Sciences

Line-based Simultaneous Localisation and Mapping (SLAM)

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What problem is this project solving?

Background

- A moving camera must determine its position while also building a map of the scene.
- RGB-D images contain both colour and depth information.
- Indoor scenes often contain many strong edges, such as desks, doors, monitors and walls.
- These edges remain visible even when there is little surface texture.

Project aim

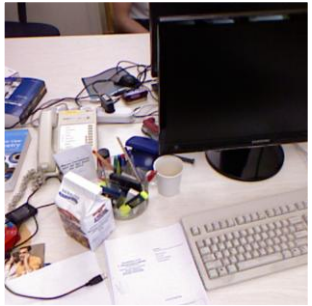
- estimate the camera path through an indoor scene
- use those poses to support 3D reconstruction from RGB-D data

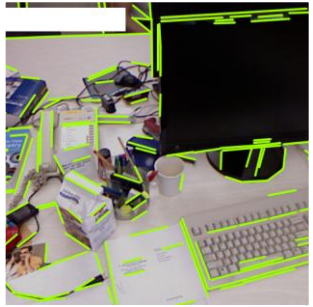
System pipeline

Line features are detected and matched between frames, incorrect matches are removed, and the remaining information is used to estimate camera motion and support 3D reconstruction.

How are useful line matches found?

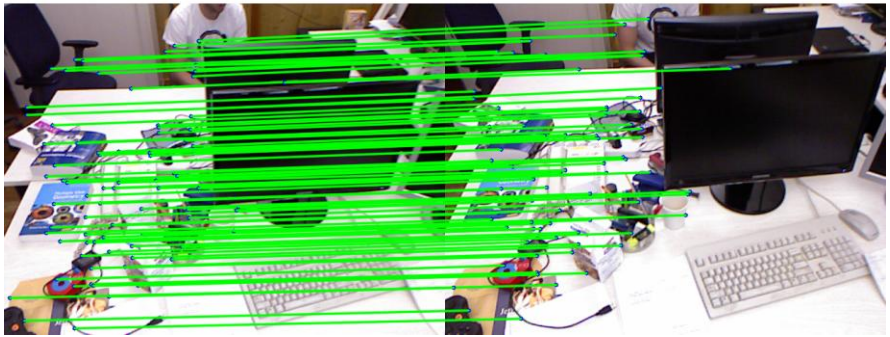
- Many lines are detected in each image.
- Some matches are correct, but many are wrong.
- A filtering step keeps only the matches that follow the main geometric pattern between the two images.
- This gives a cleaner set of matches for motion estimation.





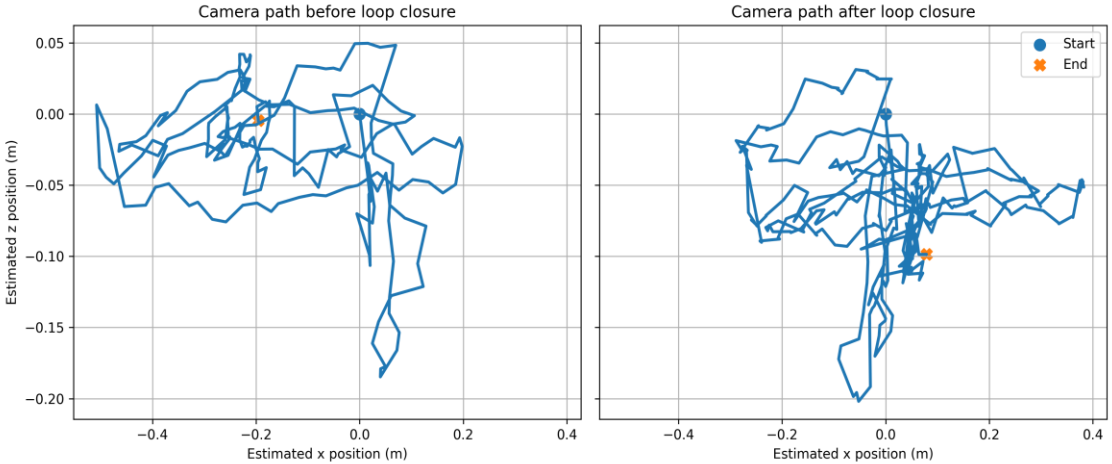
Lines detected in frame 1: 792
Lines detected in frame 2: 778
Initial matches: 515
Filtered matches: 232
Matches kept: 45.0%

Filtered line matches (232)



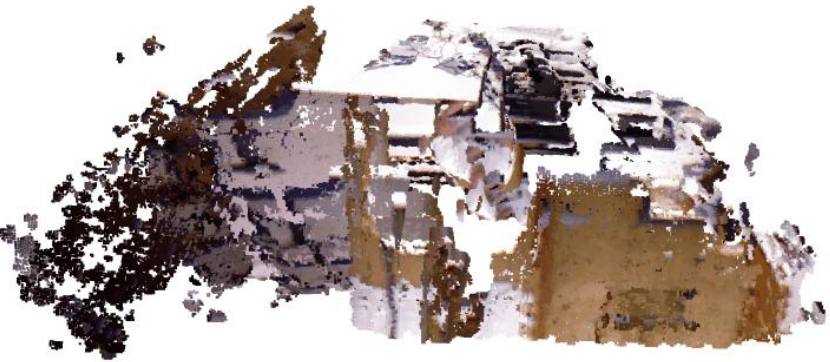
What do the tracking results show?

- The filtered line matches are used to estimate motion between one frame and the next.
- Repeating this over many frames produces a full camera path.
- Small errors build up over time, so the path can drift.
- Loop closure is used to reduce this drift when the camera returns to a place it has seen before.



Revisited places detected: 32
Revisit mismatch reduced: 0.0967 m → 0.0021 m
Average path adjustment after correction: 0.204 m

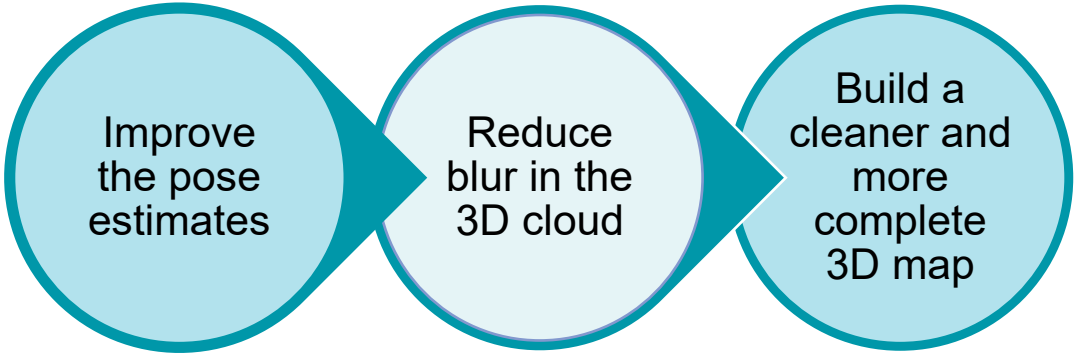
What 3D result can be produced?



- The estimated camera poses were used to place RGB-D frames into one shared 3D view.
- This produced a preliminary 3D reconstruction of the indoor scene.
- The remaining blur shows that the final quality still depends on pose accuracy.

Frames fused: 30
Depth matches found: 100%
Final point cloud size: 601,566 points

Conclusion and future work

- The project shows that line features can be used to track camera motion indoors.
- The estimated path becomes more reliable when repeated locations are used for correction.
- These camera poses can produce a multi-frame 3D point cloud.

Next steps